EXP NO: 8	MODEL EVALUATION AND IMPROVEMENT: HYPERPARAMETER TUNING WITH GRID SEARCH AND CROSS-VALIDATION

AIM:

To evaluate a machine learning model using various performance metrics and to improve its performance using feature scaling and hyperparameter tuning with GridSearchCV.

ALGORITHM:

1. Import the required libraries and load the Breast Cancer dataset.
2. Split the dataset into training and testing sets.
3. Build a baseline SVM model with default parameters on unscaled data and find its cross-validation accuracy.
4. Create a Pipeline combining StandardScaler and SVC.
5. Define a parameter grid (C, gamma, kernel) and tune the pipeline using GridSearchCV with 5-fold cross-validation.
6. Evaluate both models using accuracy, precision, recall, F1 score and ROC-AUC.
7. Plot the confusion matrices and ROC curves to compare the baseline and the improved model.

CODE:

```
# 1. Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, cross_val_score,
    GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
    f1_score,
                             roc_auc_score, roc_curve, confusion_matrix,
                                 classification_report)

# 2. Load dataset and split
data = load_breast_cancer()
X, y = data.data, data.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
```

```
def evaluate(model, name):
    pred = model.predict(X_test)
    score = model.decision_function(X_test)
    return {"Model": name,
            "Accuracy": accuracy_score(y_test, pred),
            "Precision": precision_score(y_test, pred),
            "Recall": recall_score(y_test, pred),
            "F1": f1_score(y_test, pred),
            "ROC-AUC": roc_auc_score(y_test, score)}

# 3. Baseline model: SVM with default parameters on UNSCALED data
baseline = SVC()
baseline.fit(X_train, y_train)
print("Baseline 5-fold CV accuracy:", round(cross_val_score(baseline, X_train,
    y_train, cv=5).mean(), 4))

# 4. Improvement: feature scaling + hyper-parameter tuning (GridSearchCV)
pipe = Pipeline([('scaler', StandardScaler()), ('svm', SVC())])
param_grid = {'svm__C': [0.1, 1, 10, 100],
              'svm__gamma': ['scale', 0.01, 0.001],
              'svm__kernel': ['rbf', 'linear']}
grid = GridSearchCV(pipe, param_grid, cv=5, scoring='accuracy', n_jobs=-1)
grid.fit(X_train, y_train)
print("Best parameters:", grid.best_params_)
print("Best CV accuracy:", round(grid.best_score_, 4))
best_model = grid.best_estimator_

# 5. Compare baseline and tuned models on the test set
results = pd.DataFrame([evaluate(baseline, "Baseline SVM"),
                        evaluate(best_model, "Tuned SVM (Scaled +
                            GridSearch)")])
print("\nModel Evaluation Comparison:\n")
print(results.round(4).to_string(index=False))

# 6. Detailed report of the improved model
y_pred = best_model.predict(X_test)
print("\nClassification Report (Tuned Model):\n",
      classification_report(y_test, y_pred, target_names=data.target_names))

# 7. Visualise: confusion matrices and ROC curves
fig, ax = plt.subplots(1, 3, figsize=(16, 4))
sns.heatmap(confusion_matrix(y_test, baseline.predict(X_test)), annot=True,
    fmt='d',
            cmap='Reds', ax=ax[0]); ax[0].set_title("Baseline Confusion Matrix")
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d',
            cmap='Greens', ax=ax[1]); ax[1].set_title("Tuned Confusion Matrix")
for m, n in [(baseline, "Baseline"), (best_model, "Tuned")]:
    fpr, tpr, _ = roc_curve(y_test, m.decision_function(X_test))
    ax[2].plot(fpr, tpr, label=f"{n} (AUC={roc_auc_score(y_test,
        m.decision_function(X_test)):.3f})")
ax[2].plot([0, 1], [0, 1], 'k--'); ax[2].set_xlabel("False Positive Rate")
ax[2].set_ylabel("True Positive Rate"); ax[2].set_title("ROC Curve");
    ax[2].legend()
plt.tight_layout()
plt.show()
```

OUTPUT:

```
Baseline 5-fold CV accuracy: 0.9099
Best parameters: {'svm__C': 10, 'svm__gamma': 0.01, 'svm__kernel': 'rbf'}
Best CV accuracy: 0.9802

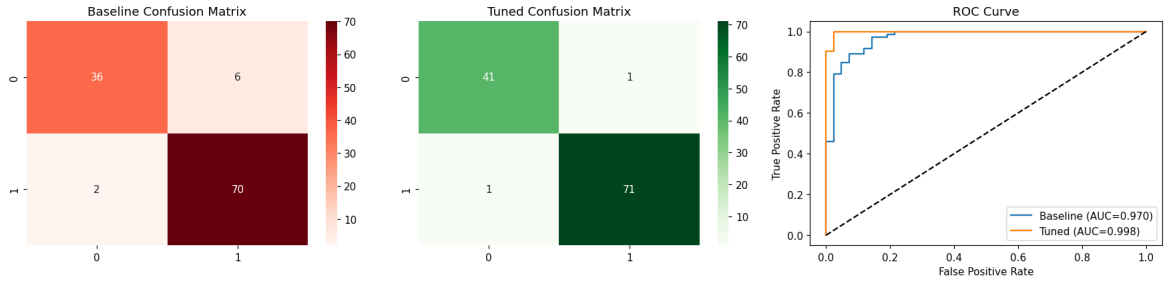
Model Evaluation Comparison:

                         Model  Accuracy  Precision  Recall     F1  ROC-AUC
                  Baseline SVM    0.9298     0.9211  0.9722 0.9459   0.9696
Tuned SVM (Scaled + GridSearch)    0.9825     0.9861  0.9861 0.9861   0.9977

Classification Report (Tuned Model):
              precision    recall  f1-score   support

   malignant       0.98      0.98      0.98        42
      benign       0.99      0.99      0.99        72

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114
```



RESULT:

The model was successfully evaluated and improved. Feature scaling and hyperparameter tuning (best parameters: C=10, gamma=0.01, kernel=rbf) increased the test accuracy from 92.98% to 98.25% and the ROC-AUC from 0.9696 to 0.9977.